

Quantum-Enhanced Agrivoltaic Digital Twin: Spectral Co-Design, Multifunctional Net Energy Benefit, and Quantum-Secured Telemetry

Steve Cabrel Tegua Kouam^{a,*}, Theodore Goumai Vedekoi^b, Jean-Pierre Tchabet Njafa^b, Jean-Pierre Nguenang^a, Serge Guy Nana Engo^b

^a*Department of Physics, Faculty of Science, University of Douala, Cameroon*

^b*Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon*

Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here we present a quantum-guided agrivoltaic system in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity acting as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.32 yr, identifying conditions under which the multifunctional net benefit stays positive for the modelled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics, quantum digital twin

*Corresponding author. Email: steve.teguia@facsciences-uy1.cm

1 Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cul-
 2 tivation to mitigate the land-use competition between food production and
 3 renewable energy generation [1, 2]. Land-use optimization and mobile-PV
 4 foundations for this field were established in *Applied Energy* [3, 4], and recent
 5 syntheses systematize overhead, vertical, and tracking typologies with land-
 6 equivalent-ratio and levelized-cost metrics [5]. Across these studies, three
 7 gaps recur and remain unresolved. (i) The optical architecture is *static*:
 8 semi-transparent PV claddings act as fixed spectral filters that attenuate ir-
 9 radiance uniformly, trading electrical output for crop biomass without opti-
 10 mization of the spectral light-sharing window [6, 7]. Dynamic, physics-guided
 11 spectral management has not been transposed into an operating agrivoltaic
 12 stack. The cost of stasis is quantifiable: in the mode-volume scan reported
 13 below, the forward transfer yield varies by a factor of 1.6 and the SERS
 14 enhancement by a factor of 48 across the accessible range (Table 2), so a
 15 statically chosen cladding fixes the energy–diagnostics balance at deploy-
 16 ment with no ability to re-trade it. The scan probes the strong-coupling
 17 penalty only: all seven volumes carry the NPoM channel, and the filtered-
 18 excitation gains quantified next apply to the NPoM-free fraction at every
 19 point of the scan. (ii) Techno-economic and lifecycle assessments remain
 20 decoupled from the underlying device physics: electricity, crop, water, and
 21 carbon accounts are evaluated subsystem by subsystem, so multifunctional
 22 trade-offs are described rather than co-optimized [8, 9]. The cost of this
 23 decoupling is concrete: accounting that ignores the local transport physics
 24 would credit the canopy with the passive baseline of 0.980 instead of the
 25 physically consistent 0.971 — and would never discover that the 91.8% local
 26 penalty costs only 0.9 points globally, the stabilizer that makes NPoM inte-
 27 gration viable. (iii) The PV layer is treated purely as an energy harvester:
 28 in-situ crop-stress and contaminant sensing is bolted on as separate hard-
 29 ware rather than co-designed with the optical stack, despite the maturity of
 30 field-deployable SERS analysis [10], of quantum-enabled agricultural work-
 31 flows [11], and of quantum-sensing readiness for precision agriculture [12].
 32 Co-design changes the economics of the sensing function: reusing the NPoM
 33 optical infrastructure delivers SERS diagnostics at a canopy-area cost of 1%
 34 and a canopy-yield cost of 0.9% relative to the passive baseline (0.980 to
 35 0.971), instead of separate sensing hardware with its own capital and foot-
 36 print.

37 The photosynthetic apparatus of green plants and bacteria has evolved
 38 under a spectrally structured photon flux, where specific wavelengths drive
 39 charge separation while others dissipate as heat [13]. This spectral specificity
 40 allows the PV layer to transmit only the photosynthetically active wave-

lengths while absorbing the remainder, converting the food-energy conflict into a spectral co-design problem.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as the model system for spectral light-sharing [14]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [15] and room temperature [16]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — the hybridization of electronic transitions with underdamped protein vibrations — governs the flow of excitation energy [17, 18]. Non-Markovian frameworks such as the hierarchical equations of motion [19] and the process tensor hierarchy of pure states (PT-HOPS) [20, 21] capture these effects with demonstrated numerical convergence. Device-adjacent uses of exact open-quantum-system methods remain rare — the closest precedent is a general HEOM library applied to light-harvesting and control problems without a photovoltaic product [22] — and photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [23, 24].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [25]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. A related study from our group, published in the *Journal of Physical Chemistry Letters*, reports the filter, excitation-transport-rate, and single-pass-conversion-efficiency analysis [26]; the present work adds the monolithic photothermal-heating hardware, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [27], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [28]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [29] that couples the biological light-harvesting sys-

tem to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and optomechanically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman-Monteith model [30] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [31] for secure IoT telemetry. This multi-scale perspective [32] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modelling framework for quantum-guided agrivoltaics (Fig. 5).

The central question of this work is falsifiable: *does an agrivoltaic stack whose spectral architecture is co-designed by exact non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (heat + power + crop + sensing) under field constraints, and which deployment barriers are physical rather than economic?* Three subordinate questions structure the Results. **(Q1) Device:** what does the NPoM co-design cost, and where does the cost land? The mode-volume scan (Table 2) quantifies the local penalty; the canopy-integration analysis shows where it is absorbed. **(Q2) Physics:** which microscopic states carry the trade-off? Exact PT-HOPS dynamics localize the loss channels, and Floquet-Stark detuning with OMIT tests whether they can be managed dynamically. **(Q3) System:** does the stack pay for itself? Coupled FAO-56 microclimate, cooperative technoeconomics, and BB84-secured telemetry internalize every function into a single Net Ecological Benefit account. Three falsifiable claim families follow. **(F1) Microscopic:** dual-band spectral filtering at 750 nm and 820 nm prepares polaron-dressed states that extend exciton coherence by 50 % and raise the forward transfer yield by 25 % at 295 K. **(F2) Device:** confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex SERS diagnostics, despite the > 90 % local transport penalty quantified across the mode-volume scan. **(F3) System:** the FAO-56 smart shield cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and, combined with OPV generation and cooperative economics, yields a Net Ecological Benefit of $72.6 \text{ kgCO}_2\text{e m}^{-2} \text{ yr}^{-1}$ and a 4.32 yr payback at 30 % subsidy for the mod-

elled 500 m² cooperative case. We answer affirmatively but conditionally: the bounds (single-exciton treatment, site-independent baths, modelled fouling and matrix penalties, $\pm 10\%$ microclimate band) are stated explicitly in the Limitations section. The headline outcome compresses this answer to one sentence, pictured in [Figure 1](#): a 91.8% local transport penalty confined to a 1% sentinel area costs 0.9 points of canopy-scale trapping yield (0.980 to 0.971) while adding single-complex SERS diagnostics to the same footprint.

Results

8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm⁻¹)**. The near-degenerate antenna pair (Sites 1–2) feeds the trap funnel (Sites 3–4) through the strongly coupled 4–5–6 backbone, and Site 8 bridges the baseplate to the trap; these couplings set the transport timescale that the spectral filter exploits. Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	−87.7	30.8	−53.5	−70.7	81.1	39.7	−15.0

We construct an 8-site model of the FMO complex, extending the standard 7-site framework [\[33\]](#) by including a coupling to the eighth bacteriochlorophyll that bridges the baseplate to the reaction center (Site 8, ~ 200 cm⁻¹ above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} = 0.15$ fs⁻¹ is the effective irreversible reaction-center trapping rate, corresponding to a capture time $\tau_{\text{trap}} \approx 6.7$ fs — short compared with the 1-ps propagation window, so that near-complete capture (the 0.98 passive baseline) is dynamically accessible in [Eq. \(4\)](#) [\[34\]](#). The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. All 28 inter-site couplings are validated to be finite, and the Hamiltonian is verified to be non-Hermitian only on the trapping diagonal terms ([Section S1](#)).

140 *NPoM plasmonic strong coupling*

141 The OPV-FMO interface is mediated by a nanoparticle-on-mirror
 142 (NPoM) architecture [29], where a gold nanoparticle (diameter 80 nm) is
 143 separated from a gold mirror by a 1 nm graphene monolayer spacer. The
 144 graphene barrier prevents Dexter-type electron transfer while maintaining
 145 electromagnetic field enhancement. The NPoM cavity supports an intense
 146 localized surface plasmon mode that couples coherently to the FMO optical
 147 transitions and mediates the optical interface with the OPV layer.

148 The single-emitter coupling strength follows the cavity quantum electro-
 149 dynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

150 with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [29]. At the sub-nanometer mode vol-
 151 umes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1}
 152 to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the sys-
 153 tem in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plas-
 154 mon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$
 155 to prevent divergences. Recent advances in quantum-tunnel-junction plas-
 156 monic sources have demonstrated that such sub-nanometer cavities can be
 157 self-illuminating, eliminating the need for bulky external optics [35].

158 The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

159 where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the
 160 primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength
 161 g_0 . Figure 2a shows the energy level diagram.

162 *Non-Markovian quantum dynamics*

163 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
 164 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
 165 terms, time step $\Delta t = 0.2 \text{ fs}$, and $n = 100$ disorder realizations ($\sigma = 50 \text{ cm}^{-1}$
 166 static disorder) [20, 21]. The bath is modeled as a Drude-Lorentz over-
 167 damped component ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) plus 12 underdamped
 168 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
 169 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara

truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$ and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-aware scheduling for these high-order simulations are detailed and validated in our companion study [25].

Figure 2b shows the exciton population dynamics under broadband *vs.* dual-band filtered excitation. The filtered case (solid lines) maintains higher population on the trapping sites (Sites 3 and 4), leading to enhanced forward transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for broadband excitation — a relative enhancement of 25 % (Figure 2c). These filtered- and broadband-excitation simulations of the bare 8-site FMO Hamiltonian (no NPoM channel) are inherited unchanged from our companion study, where their full trajectory sets were reported [25]; they characterize the 99 % passive fraction of the canopy and are therefore independent of, and complementary to, the NPoM mode-volume scan below, which probes the strong-coupling penalty on the sentinel fraction. The coherence lifetime τ_c , defined as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [36], extends from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase that is causally linked to the yield gain: the extended coherence suppresses the pure-dephasing rate, lengthening the effective transport mean-free path and thereby increasing the fraction of the excitation wavepacket that reaches the reaction center before decohering. The mechanism is identical to that established for the 7-site system [25]: the dual-band filter selectively populates polaron-dressed states whose residual reorganization energy λ_{res} is reduced by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

NPoM cavity volume dependence of exciton transport

The nanoparticle-on-mirror cavity couples the FMO complex to the OPV active layer through the strong coherent interaction described above, but the plasmonic mode also acts as an additional population decay channel that competes with the reaction center trapping. To quantify this trade-off, we perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls

the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller mode volumes produce stronger plasmon–exciton coupling but also enhance the plasmon’s role as a population sink, diverting excitation away from the reaction center.

Table 2: NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement. Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [29]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 3	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 1	−91.9	33

Table 2 summarizes the forward transfer yield across seven mode volumes spanning the experimentally accessible sub-nanometer range [29]. The yield increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} = 0.2 \text{ nm}^3$ to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, with a slight decrease at 1.4 nm^3 . The trend is logarithmic with diminishing returns above 1.0 nm^3 ; increasing the volume beyond 1.0 nm^3 recovers only 1% of the baseline yield.

The physical origin of this suppression is the strong-coupling-induced hybridization of excitonic and plasmonic states. The dressed eigenstates acquire a finite plasmonic character, which decays via the fast plasmon relaxation pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center. At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting narrows, and the excitonic character of the bright states is partially restored.

223 Despite the monotonic recovery, all NPoM-on yields remain suppressed
 224 by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This
 225 indicates that, in the strong collective coupling regime accessible to NPoM
 226 cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally
 227 limits the maximum achievable transport efficiency. The SERS enhancement
 228 factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$, following
 229 the Purcell enhancement of the local field at both the excitation and emission
 230 wavelengths. [Table 2](#) includes the calculated SERS enhancement for each
 231 mode volume, revealing a four-order-of-magnitude range (33 to 1.6×10^3)
 232 across the scanned volumes. The optimal volume represents a compromise:
 233 $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while retaining
 234 SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1} Franck–
 235 Condon mode populations at the single-FMO-complex level [10]. This com-
 236 promise only works because of the sentinel architecture — no volume on the
 237 scan simultaneously restores transport and maximizes diagnostics, which is
 238 precisely why the 1 % area split, not a better volume, is the design lever.
 239 By contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the
 240 cost of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to
 241 diagnostic-only applications where photochemical efficiency is secondary. At
 242 the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$
 243 confirms the same limiting behaviour ($\Phi_{\text{FT}} = 0.0768$, final plasmon popula-
 244 tion 0.79; [Section S11](#)).

245 Second, reducing the temperature to 77 K (analyzed as a theoretical limit-
 246 case for thermal phonon suppression in [Section S11.2](#)) doubles the trapping
 247 yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K,
 248 the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for
 249 $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass
 250 plasmonic loading.

251 All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho =$
 252 1.0000), validating the numerical stability of the PT-HOPS/SBD propagator.
 253 The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations;
 254 the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} =$
 255 0.0799 , within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off
 256 run ($n = 20$) is shown as a limiting-case reference; quantitative comparison
 257 across configurations is limited by the differing trajectory counts.

258 *Global canopy yield integration*

259 Because the NPoM SERS diagnostic mode suppresses local excitonic
 260 transport through plasmon population loading, it need only cover a 1 % sen-
 261 tinel fraction α of the canopy area. The remaining 99 % operates in passive
 262 OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} =$
 263 0.980. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

264 yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 pre-
 265 cisely); canopy-scale productivity is thus maintained even though transport
 266 is suppressed locally.

267 *Dynamic photo-protection via Floquet–Stark detuning and OMIT*

268 Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant
 269 changes dynamically, inducing a time-dependent Stark shift of the excitonic
 270 levels that detunes the OPV–FMO coupling. We model this as a Floquet
 271 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

272 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_{\text{F}} = 1.2 \text{ THz}$. When the in-
 273 cident flux exceeds the threshold, the time-dependent detuning shifts the
 274 OPV exciton energies out of resonance with the FMO absorption bands, re-
 275 ducing energy transfer to the biological complex by up to 40 % (Figure 3a).
 276 To prevent rapid, noise-induced oscillations under fluctuating cloud cover,
 277 a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning
 278 switch at 850 W m^{-2} and deactivating it only when the flux drops below
 279 750 W m^{-2}).

280 Concurrently, an optomechanically induced transparency (OMIT) mecha-
 281 nism [37] in the NPoM cavity creates destructive interference that suppresses
 282 transmission at the 750 nm and 820 nm channels under high flux. The trans-
 283 mission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

284 where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At 1000 W m^{-2} , the trans-
 285 mission drops to 44 % of baseline (Figure 3b), providing dynamic protection
 286 against exciton overload. At zero flux (night mode), both mechanisms return
 287 to zero without singularities (Section S4).

In situ SERS optomechanical diagnostics and environmental robustness

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39], which is the integration gap addressed here. Multilayer graphene-metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [40]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode volumes.

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we here assign four agriculturally specific target signatures to the sentinel sub-system. **Oxidative stress** is read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} , with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [10]. **Pesticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its characteristic Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$ under controlled conditions), leveraging the plasmonic hot-spot amplification in the NPoM gap [10]. **Heavy metal contamination** of irrigation water (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encapsulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$ under controlled conditions) [41] sealed within the NPoM sensor probe polymer matrix, which are immune to the dust and biofouling that affect bare gold surfaces. Cross-sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual spectral interrogation: heavy metals modify the Stern–Volmer quenching intensity and fluorescence decay time, whereas soil moisture alters the local dielectric constant, producing a spectral broadening that is spectrally orthogonal to the metal-induced quenching signature. **Pathogen detection** is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at the greenhouse hub (Section S12.7). These solid-state quantum sensors detect paramagnetic bacterial metabolites from sap or irrigation water samples at pre-symptomatic stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$ under controlled conditions; the agricultural matrix will degrade in-field sensitivity, which is mitigated by the graphene barrier and *DynamicCalibrator* corrections).

Figure 3c shows the simulated Raman spectrum for the three BChl *a* modes under active NPoM coupling. The relative intensities encode the in-

stantaneous exciton population distribution, providing a direct optical probe of the energy transfer pathway.

Environmental robustness. The realistic agricultural environment (dust, bird droppings, microbial fouling) is modeled at three infrastructure levels. At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning interval. At the sensor level, the graphene monolayer spacer of each NPoM cavity is functionalized with a zwitterionic polymer coating [42, 43], modeled in our simulations to preserve ≥ 90 % quantum coherence after 60 d continuous field immersion; this coating constitutes the primary anti-fouling barrier that physically repels microbial adhesion and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and CQD sensor tip further attenuates the diurnal temperature excursion experienced by the quantum probe, attenuating the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$ (Section S12.5). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety net: it applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

Smart shield microclimate: FAO-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. Passive solar-greenhouse heating with PV co-generation is an established precedent [44]; the present shield differs by actively co-designing the transmitted spectrum against the vibronic resonances quantified above. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [30] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the

psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25°C , 60% humidity, 2 m s^{-1} wind), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28% reduction — translating to 1.3 mm d^{-1} in water savings (Figure 4a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [30].

Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$), which reduces the entry-level capital expenditure to $161\,000 \text{ USD}$ (at 322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the Monte-Carlo techno-economic logic established for European agrivoltaic configurations [45]. A blended-finance subsidy of 30% reduces the net CAPEX to $112\,700 \text{ USD}$.

Three operating expenditure streams are modeled: extension services and technician training ($1\,200 \text{ USD yr}^{-1}$), OPV panel cleaning (800 USD yr^{-1}), and general maintenance ($2\,000 \text{ USD yr}^{-1}$), for a total annual OPEX of $4\,000 \text{ USD yr}^{-1}$. The net annual cash flow of $26\,090 \text{ USD yr}^{-1}$ (from $30\,090 \text{ USD yr}^{-1}$ total revenue minus $4\,000 \text{ USD}$ of OPEX) yields a simple payback period of 4.32 yr (or 6.17 yr without subsidy) and a 10-year discounted NPV of $+34\,714 \text{ USD}$ at a 12% risk-adjusted discount rate appropriate for Sub-Saharan Africa [46] (Figure 4c). Holding other inputs fixed, varying the subsidized net CAPEX by $\pm 20\%$ shifts the payback to $3.46\text{--}5.18 \text{ yr}$; a 50% higher OPEX ($6\,000 \text{ USD yr}^{-1}$) gives 4.49 yr , and 20% lower revenue gives 5.61 yr . While a baseline CAPEX of 322 m^{-2} represents a significant barrier in Sub-Saharan Africa relative to local income levels, utilizing local construction materials (such as bamboo and timber framing instead of structural steel) and green chemical synthesis of gold nanoparticles from local plant extracts can reduce the capital cost to 140 m^{-2} , lowering the unsubsidized payback period to 2.68 yr and making the modelled system potentially viable for local cooperatives under these cost assumptions. By contrast, the remaining barriers are physical rather than economic: fabrication of sub- 2 nm NPoM gaps still requires laboratory-grade nanofabrication and a stable Au/graphene supply chain, and OPV encapsulation lifetime under tropical irradiance is the binding reliability constraint (Section S12.5). Subsidy policy alone cannot clear these barriers;

they define the process-engineering roadmap that separates this modelled proof of concept from field deployment. Taken together, these sensitivities demonstrate that high-value crop selection and blended-finance structurally transform the investment case.

Secure IoT telemetry via BB84 quantum key distribution

Field-to-cloud telemetry is secured using a simulated BB84 quantum key distribution protocol [31] over a buried single-mode fiber link, which eliminates the atmospheric turbulence and scintillation that affect free-space channels in agricultural environments. This quantum-secured telemetry layer is physically integrated with the NPoM-CQD sentinel network: because the high-sensitivity biological and pesticide diagnostics control high-stakes irrigation and crop-protection actuators (which could be targeted by hostile actors to cause artificial crop failure or pesticide contamination), securing these feedback commands with BB84 QKD protects the command integrity of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [47]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a sample fraction; if QBER exceeds the Shor-Preskill unconditional-security threshold of 11 % [48], the protocol raises a **SecurityThresholdExceeded** exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [47], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

Experimental testability

The modelled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First,

431 two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-
 432 modulator pulse shaping into dual-band channels at 750 nm and 820 nm
 433 can track the diagonal and cross-peak quantum-beat oscillations predicted
 434 here, verifying the extension of τ_c from 280 fs to 420 fs [49, 16]. Second,
 435 time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1}
 436 and 1145 cm^{-1} BChl *a* modes in real time and quantify the area-weighted
 437 yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [29, 50]. More broadly,
 438 recent assessments identify quantum sensing for precision agriculture as an
 439 emerging opportunity whose open calibration and robustness gaps match
 440 the calibration architecture proposed here [12].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295\text{ K}$, $L = 8$, $K = 2$.

Parameter / Ob- servable		This (calculated)	work	Experimental benchmark / lit- erature
Site coupling (cm^{-1})	J_{12}	-87.7		-87.7 ± 5.0 (2DES, Brixner 2005 [49])
Reorganization energy λ_D (cm^{-1})		35		35–40 (2DES, Engel 2007 [15])
Cutoff frequency (cm^{-1})	γ_D	50		50 (2DES, Panitchayangkoon 2010 [16])
Coherence lifetime (passive)	τ_c	$420 \pm 35\text{ fs}$		$400 \pm 50\text{ fs}$ (2DES room temp., Pan- itchayangkoon 2010 [16])
NPoM coupling ($V_{\text{mode}} = 1.2\text{ nm}^3$)	g_0	110 cm^{-1}		$120 \pm 15\text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [29])
SERS EF _{SERS}	enhancement	44		10^2 – 10^3 (sub-nm NPoM, Baumberg 2019 [50])

441 Discussion

442 Our results demonstrate that quantum coherence engineering at the bio-
 443 organic interface can deliver modelled sustainability benefits across micro-
 444 scopic energy transfer, crop water use, carbon footprint, and data security.

445 **Mechanism generality.** The dual-band filtering effect is not specific
 446 to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Sec-
 447 tion S7) is a general property of vibronically coupled chromophore aggre-
 448 gates [17]. We verified this by applying the protocol to a minimal three-site

excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [28]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection. Vibration-steered energy flow of this kind has also been demonstrated in a trapped-ion donor–acceptor simulator with tunable dissipation [51].

The “quantum divide.” The advanced materials required for quantum agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet accessible to subsistence farmers in developing economies. Our cooperative cost-sharing model demonstrates that with a carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback period drops below 4 yr, making the technology economically viable under the modelled subsidy. We propose a tiered subsidy framework where the NEB itself serves as the verifiable metric for subsidy qualification, linking quantum yield improvements to farmer access.

Quantum digital twin. The modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a model-based supervisory replica that combines sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to recommend irrigation schedules, OPV cleaning dates (from soiling factor Eq. (8)), and carbon credit yields. This device-physics integration is distinct from the supply-chain and breeding applications mapped in recent quantum-in-agriculture reviews [11].

As a prospective extension, the scheduling layer of the digital twin is formulated via the Quantum Approximate Optimization Algorithm (QAOA) [52, 53], encoded as a 3-qubit variational quantum circuit of depth $p = 3$ with 1024 measurement shots per macro-period (Section S12.3). The algorithm encodes the irrigation–refrigeration–grid-export trilemma as a quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and solves it via a PennyLane emulator at sub-hourly timescales. Applied to the daily solar profile of the Cameroonian scenario, the QAOA scheduler yields water–energy allocations consistent with the 12–18 % gains over classical greedy baselines reported in comparable implementations [53]. As a further prospective readout, our `quantum_gravimetry.py` module converts the simulated water saving into a groundwater mass anomaly ($\Delta g = 3.5 \text{ nm s}^{-2}$ for a single greenhouse — below single-instrument noise but potentially detectable

487 with a small regional gravimeter network or gradiometric common-mode
 488 rejection), with the full noise-budget and inversion discussion given in
 489 [Section S12.2](#) [54, 55].

490 A hybrid quantum-classical signal processing pipeline — combining
 491 Matrix Product State (MPS) denoising with quantum kernel ridge re-
 492 gression (QKRR) [56, 57] — further enables the digital twin to extract
 493 pre-symptomatic stress signatures from the noisy SERS and CQD fluores-
 494 cence telemetry. To satisfy edge computation constraints, the heavy MPS
 495 tensor contraction and QAOA optimization are executed on the centralized
 496 cooperative server (or cloud), with the resulting optimized weights transmit-
 497 ted back to the field nodes — an edge-cloud split complementary to classical
 498 drift-diffusion device simulators [58]. The local ARM Cortex-M4 IoT node
 499 executes only the low-latency inference of the pre-trained, frozen-weight
 500 QKRR model. The MPS compressor acts as a nonlinear low-pass filter on
 501 the server that suppresses greenhouse noise (thermal fluctuations, biofoul-
 502 ing, dust) while preserving the sharp vibronic features at 180 cm^{-1} and
 503 1145 cm^{-1} that classical Savitzky–Golay filters tend to smear ([Section S10](#)).
 504 The cleaned spectra are projected through the quantum-inspired kernel for
 505 anomaly scoring, issuing early warnings that allow preemptive agronomic
 506 intervention before macroscopic symptoms appear — all within a 3.7 ms local
 507 compute budget on the IoT node. We emphasize that this sub-millisecond
 508 execution latency is not designed to match the slower multi-hour physical
 509 response times of soil moisture percolation and plant transpiration. Rather,
 510 it enables spatial multiplexing: a single ARM Cortex-M4 edge processor
 511 can sequentially query, compress, and classify SERS telemetry from dozens
 512 of distributed NPoM sensor nodes across the 500 m^2 greenhouse in under
 513 a second, reducing the electronic-component count. This chain connects
 514 microscopic quantum efficiency to field-level agronomic management; no
 515 existing agrivoltaic design integrates both [46, 6, 2].

516 **Prior art and positioning.** Previous agrivoltaic studies have focused
 517 on light-sharing through passive spectral filtering [6] or crop selection [2].
 518 Semi-transparent OPVs with power conversion efficiencies exceeding 15 %
 519 and 40 % average visible transmittance have recently been demonstrated
 520 through blade-coating scale-up methods [59], and complementary work on
 521 semitransparent organic photovoltaics for greenhouse applications has shown
 522 power conversion efficiencies above 12 % [7]. Our contribution is to add
 523 an active, quantum-coherent dimension to the spectral filter: the OPV is
 524 no longer a passive absorber but an active spectral shaper that prepares

coherence-protected states in the photosynthetic apparatus. Multi-vector solar agri-systems combining agrivoltaics with hydrogen and desalination set the *Applied Energy* bar for water–food–energy integration [60]; we differentiate by quantum-guided spectral co-design with integrated sensing, and we adopt the functional-unit discipline (emissions per kWh, economics per configuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this literature, the present work provides, to our knowledge, the first (i) agrivoltaic stack whose spectral architecture is co-designed by numerically exact open-quantum-system dynamics, jointly optimizing excitation transport and power conversion; (ii) monolithic photothermal–PV–SERS device concept with field-relevant calibration in a single footprint; and (iii) multifunctional techno-economic and lifecycle framework that internalizes heat, crop, water, sensing, and data-security trade-offs in $\$/W + \$/kg + \$/m^3$ terms. Table 4 condenses this positioning: each prior class delivers two to three of the seven functions, whereas the present stack co-optimizes all seven in a single footprint under one accounting.

Table 4: **Capability matrix of agrivoltaic and agri-sensing approaches.** ✓ = function delivered within the stack; – = partial or incidental; — = absent. Classes: passive spectral claddings [6, 2, 7]; multi-vector agri-energy systems (agrivoltaics coupled to hydrogen and desalination) [60]; standalone agri-sensing hardware [10, 11, 12]. Multifunctional accounting denotes joint $\$/W + \$/kg + \$/m^3$ techno-economic and lifecycle evaluation of all functions. No prior class delivers more than three of the seven functions; the present stack delivers all seven in a single footprint with unified accounting.

Function	Passive	Multi-vector	Sensing	This work
Electrical power (OPV)	✓	✓	—	✓
Crop microclimate	✓	✓	—	✓
Water management	–	✓	—	✓
Photothermal heat	—	–	—	✓
In-situ SERS sensing	—	—	✓	✓
QKD-secured telemetry	—	—	—	✓
Multifunctional accounting	—	—	—	✓

This approach is distinct from prior quantum biology studies [61] that treat the environment as a fixed noise source; here, the photonic environment itself is engineered through the NPoM cavity to maximize coherence protection. A key design choice warrants explicit justification: the NPoM cavity, despite imposing a $> 90\%$ penalty on excitonic transport (Table 2),

546 is retained because its *diagnostic function* is dedicated — it enables non-
547 destructive, high-sensitivity SERS readout of the FMO vibrational state,
548 pesticide levels, and heavy-metal contamination. The transport penalty is
549 absorbed by confining NPoM operation to the 1 % sentinel fraction ([Equa-
550 tion \(5\)](#)), preserving 97.1 % of the global canopy trapping yield while main-
551 taining full diagnostic capability.

552 **Limitations.** Several approximations constrain our analysis. The quan-
553 tum dynamics are restricted to the single-exciton manifold; under high solar
554 flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in
555 the OPV and FMO interfaces could alter the collective coupling strength
556 g_{eff} and transport efficiency. The bath is treated as site-independent, and
557 the reaction center trapping is modeled as a phenomenological Lindblad-like
558 rate. The FAO-56 model, while standard, does not capture transient stom-
559 atal responses to fluctuating light or the dynamic microclimatic feedback
560 between canopy transpiration and vapor pressure deficit (VPD) inside the
561 greenhouse envelope. Sensitivity analysis over plausible wind-speed varia-
562 tions ($u = 0.2 - 0.8 \text{ m s}^{-1}$) yields a $\pm 10 \%$ uncertainty band on the predicted
563 28 % water saving, which propagates into a $\pm 10 \%$ range on the annual irri-
564 gation volume and corresponding NEB carbon offset. The BB84 simulation
565 assumes a buried single-mode fiber link; while buried fibers are vulnerable to
566 agricultural mechanical damage (plowing, root intrusion), a free-space opti-
567 cal (FSO) link would suffer from strong scattering under dusty greenhouse
568 environments, requiring dedicated atmospheric correction models. While
569 software-based post-quantum cryptography offers a lower-cost alternative,
570 the physical BB84 protocol provides information-theoretic security immune
571 to algorithmic cryptanalysis on resource-constrained embedded nodes such
572 as the ARM Cortex-M4 irrigation controllers. The 1 % sentinel fraction is a
573 design parameter; field optimization will require empirical calibration of the
574 SERS LODs under actual fouling conditions. We conservatively estimate a
575 $\times 10$ matrix-effect penalty on all reported LODs when transitioning from con-
576 trolled laboratory buffers to raw agricultural sap or irrigation water, owing
577 to competitive binding, background fluorescence, and ionic strength effects;
578 the *DynamicCalibrator* compensates for this attenuation through real-time
579 baseline correction. The zwitterionic polymer coating modeled on the NPoM
580 graphene spacer ([Section S12.1](#)) acts as the primary fouling barrier in our
581 simulations, with the *DynamicCalibrator* serving as a secondary correction
582 layer; the dual architecture reduces the mandatory cleaning frequency, which
583 is already factored into the cooperative OPEX model. We stress that the 60-

584 day resilience of zwitterionic coatings applies strictly to sensors sheltered in
585 protected micro-sampling chambers or rigid ceramic encasings; direct, un-
586 protected exposure to abrasive soil matrices (bioturbation, quartz particles,
587 tillage) would lead to mechanical scraping and rapid surface degradation.

588 The 30 % blended-finance subsidy required to achieve a 4.32yr payback
589 period cannot be bridged by voluntary carbon credits alone (generating only
590 363 USD yr⁻¹, about 1.4 % of the net cash flow, at 10 USD t_{CO₂e}⁻¹ for the
591 500 m² module); it must be underpinned by dedicated rural-adaptation fi-
592 nance channels such as the Green Climate Fund (GCF) and the African
593 Development Bank’s Climate Change Fund (Fonds Spécial Climat-Dev de
594 la BAD), which are mandated to co-finance climate-resilient agriculture in
595 Sub-Saharan Africa.

596 **Outlook.** The integration of quantum coherence engineering with agri-
597 cultural sustainability suggests an axis for agrivoltaic design. Three experi-
598 mental and theoretical directions are prioritized for future work.

599 First, on the hardware and materials scale, the quantum model must
600 be extended to full multi-exciton dynamics under realistic solar spectra, ac-
601 companied by experimental validation via two-dimensional electronic spec-
602 troscopy (2DES) with spatial light modulator pulse shaping [62, 63]. Field tri-
603 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
604 sentinel network are required to evaluate in-situ sensor performance under
605 actual biofouling conditions and verify the coherence-preservation lifetime
606 of the zwitterionic functionalization layer (Section S12.1). Alternative low-
607 loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) support-
608 ing Mie resonances [64] and nanocavity-free substrates — should be explored
609 to minimize the NPoM transport penalty while preserving SERS sensitivity.
610 Additional hardware extensions include passive thermal micro-shielding (Sec-
611 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent
612 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-
613 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

614 Second, scaling the digital twin to a multi-cooperative network will re-
615 quire hardware-in-the-loop QKD validation under dusty field conditions and
616 deployment of a real-time aquifer gravimetric array to confirm that the mod-
617 eled groundwater recharge connects at the satellite geodesy scale. Monte
618 Carlo uncertainty propagation in the full LCA should be extended with cor-
619 related climatic input distributions drawn from regional reanalysis data.

620 Third, to prevent these advances from widening the “quantum divide,”
621 we have defined the Global Quantum Agriculture Standard (GQAS), a six-

pillar certification framework covering QKD security, data sovereignty, sensor detection limits, quantum fidelity, latency, and lifecycle sustainability (Section S12.9). Our Cameroon cooperative proof-of-concept scenario serves as the first application of this proposed framework, satisfying all six GQAS criteria and demonstrating internal consistency across the multi-domain simulation. Independent experimental validation will be required before GQAS can be adopted as an operational standard. Scaling GQAS adoption to small-holder networks across Sub-Saharan Africa and Southeast Asia requires subsidized sensor supply chains, open-access certification auditing, and regulatory recognition on premium export markets such as GlobalGAP.

Conclusions

We asked whether exact non-Markovian quantum dynamics can jointly optimize excitation-transport and power-conversion efficiency in a sensing-integrated photothermal agrivoltaic stack, with a positive multifunctional net benefit under field constraints. Within the modelled 500 m² cooperative case, the answer is affirmative but conditional: dual-band filtering raises the forward transfer yield by 25 % and extends coherence lifetime by 50 %; confining NPoM operation to a 1 % sentinel fraction preserves 97.1 % global canopy yield while retaining SERS diagnostics; the FAO-56 smart shield cuts evapotranspiration by 28 % with a Net Ecological Benefit of 72.6 kgCO_{2e} m⁻² yr⁻¹; and the cooperative case pays back in 4.32 yr at 30 % subsidy. These claims are bounded by the > 90 % local NPoM transport penalty, the single-exciton and site-independent-bath approximations, the ±10 % microclimate uncertainty band, and the modelled (not field-validated) fouling and matrix-penalty assumptions. For deployment, the economic barrier is dominant and tractable — local-material construction and green-synthesis gold lower the capital cost from 322 USD m⁻² to 140 USD m⁻² (unsubsidized payback 2.68 yr) — whereas the physical barriers (nanofabrication of sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV encapsulation lifetime under tropical irradiance) require process engineering rather than subsidies. Priority next steps are 2DES validation of the filter protocol, low-loss dielectric antenna alternatives, field fouling trials of the sentinel network, and a localized Au/graphene sensor supply chain for Sub-Saharan manufacturing.

656 Methods

657 *Proof-of-Concept scenario: 500 m² Floriculture Micro-Module*

658 To ground the multi-domain framework in a concrete operational context,
 659 we consider a 500 m² high-tunnel greenhouse located in Cameroon’s West-
 660 ern Highlands. This region supports high-value floriculture (strawberries, cut
 661 flowers) suitable for micro-module deployment by cooperatives. The environ-
 662 ment features a mean annual solar irradiance of 5.76 kWh m⁻² d⁻¹ [65] and
 663 significant biofouling potential that necessitates the NPoM *DynamicCalibra-*
 664 *tor* and regular OPV cleaning (factored into the cooperative OPEX model).
 665 Under the NPoM-shield spectral filter, the 28 % reduction in crop evapo-
 666 transpiration (ET_c) relative to the open-field FAO-56 baseline corresponds
 667 to the modelled water-savings credit of 474.5 L m⁻² yr⁻¹ (237 m³ yr⁻¹ for the
 668 500 m² greenhouse; Section S5). The area-weighted global canopy trapping
 669 yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodat-
 670 ing a 1 % NPoM sentinel fraction for SERS diagnostics. Crop production is
 671 assumed at the horticultural standard for the crop mix (8.2 kg m⁻² yr⁻¹); the
 672 quantum-fertiliser channel adds 0.96 kg m⁻² yr⁻¹ (8 % biostimulation, Sce-
 673 nario A only), and no coherence-derived yield gain is monetized — the
 674 passive-fraction coherence gains of Figure 2c are treated as a device-physics
 675 result, not a crop-revenue stream. The OPV module generates 90 MWh yr⁻¹
 676 of electrical energy for a 500 m² installation at 180 kWh m⁻² yr⁻¹ — the
 677 guaranteed-yield accounting (5.76 kWh m⁻² d⁻¹ irradiance, 15 % module ef-
 678 ficiency, 0.57 performance ratio) that is used consistently in both the rev-
 679 enue and the carbon accounts. Displacing grid electricity at 450 g_{CO_{2e}} kWh⁻¹
 680 offsets 40.5 t_{CO_{2e}} yr⁻¹ (81 kg_{CO_{2e}} m⁻² yr⁻¹). Combined with the NEB (72.6
 681 kg_{CO_{2e}} m⁻² yr⁻¹, Figure 4b), the system achieves net-zero operational car-
 682 bon rapidly. At 10 USD t_{CO_{2e}}⁻¹ [66], the system generates 363 USD yr⁻¹ in
 683 verified carbon revenue. The 500 m² scale is well matched to cooperative
 684 ownership models: under a 30 % blended-finance subsidy, the payback short-
 685 ens from 6.17 yr unsubsidized to 4.32 yr, with each of the five cooperative
 686 members exposed to only 22 540 USD of net capital. This demonstrates that
 687 quantum-enhanced agrivoltaics can be economically viable under modelled
 688 assumptions when coupled with strategic crop selection and robust field-
 689 sensor integration.

690 *PT-HOPS/SBD quantum dynamics*

691 All quantum dynamics simulations use the Process Tensor Hierarchy of
 692 Pure States with Stochastically Bundled Dissipators as implemented in the
 693 open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$
 694 with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2$ fs, established as the
 695 production standard for converged dynamics at 295 K [25]. Static disorder
 696 is modeled as Gaussian fluctuations of the site energies with standard devi-
 697 ation $\sigma = 50$ cm $^{-1}$, sampled over $n = 100$ realizations. The bath spectral
 698 density follows the 12-mode Kleinekathöfer/Coker parameterization with a
 699 Drude–Lorentz background ($\lambda_D = 35$ cm $^{-1}$, $\gamma_D = 50$ cm $^{-1}$) and 12 under-
 700 damped vibronic modes (frequencies 180–1 500 cm $^{-1}$, Huang–Rhys factors
 701 $S_k = 0.003 - -0.050$).

702 *FAO-56 evapotranspiration*

703 Crop evapotranspiration is computed using the standard FAO-56
 704 Penman–Monteith equation [30] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

705 with R_n attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$. We note that
 706 the modified FAO-56 formulation represents a first-order approximation; un-
 707 der the enclosed high-tunnel envelope, boundary layer resistance and sensible
 708 heat exchange are altered relative to open field conditions. Future iterations
 709 of the digital twin should integrate three-dimensional Computational Fluid
 710 Dynamics (CFD) modeling to map localized microclimatic gradients and la-
 711 tent heat transfer. Wind speed within the greenhouse is modeled at a static
 712 representative value of $u = 0.5$ m s $^{-1}$, corresponding to a typical 10 % reduc-
 713 tion of average external wind velocities. Temperature is fixed at 25 °C and
 714 relative humidity at 60 % unless specified otherwise.

715 *BB84 QKD simulation and fail-safe IoT architecture*

716 The BB84 protocol is implemented as a Python simulation of polarization-
 717 state preparation, noisy transmission over a buried single-mode fiber
 718 channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random
 719 bits (for $N = 256$ key length) in random rectilinear or diagonal bases.
 720 Bob measures in random bases, and matching-basis events are retained
 721 for key sifting. QBER is estimated on a sample fraction (minimum 1

bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of 11 % [48], a `SecurityThresholdExceeded` exception is raised, halting the QKD-protected irrigation command relay. A last-validated soil-moisture reading from the *DynamicCalibrator* is used to execute a conservative local irrigation fallback.

SERS enhancement scaling limits

The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode volume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of the localized electromagnetic field. We note that this standard $|E|^4$ approximation represents a lower-bound semi-classical estimate. In sub-nanometer picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling, and non-local dielectric screening alter the field distribution relative to classical electrodynamics, while molecular optomechanical coupling can pump vibrational modes out of thermal equilibrium. Precise calculation of the localized fields in this regime requires non-local Green’s tensor integration, which motivates the proposed in-situ experimental validation. The benchmark against the experimentally reported 10^2 – 10^3 range for sub-nanometer NPoM gaps [50] is given in Table 3.

DynamicCalibrator, thermal shielding, and soiling factor

Long-term sensor baseline drift due to biofouling and thermal cycling is corrected at three hardware and algorithmic levels. The primary fouling barrier is the zwitterionic polymer functionalization layer deposited on the graphene monolayer spacer of each NPoM cavity [42, 43], which physically repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coherence after 60 d field immersion. As a secondary hardware barrier, a thin-film micro-shielding layer ($1 \mu\text{m}$ to $5 \mu\text{m}$ of SiO_2 or Al_2O_3) is deposited around the NPoM picocavity and CQD sensor tip to insulate the quantum probe from diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the probe experiences only 15 % of the ambient greenhouse temperature fluctuation, reducing the thermal contribution to Stern-Volmer drift from 5.25 % to 0.79 % under a $\pm 15 \text{ K}$ diurnal cycle. The *DynamicCalibrator* acts as a tertiary algorithmic correction: an exponential moving average (EMA) with smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the EMA $\hat{\mu}(t)$ exceeds 20 % of the initial baseline, triggering OPV panel cleaning.

757 At the panel level, OPV electrical output is attenuated according to
758 Eq. (8), with a default decay rate of $0.5\% \text{ d}^{-1}$ and a minimum soiling factor
759 of 0.75 (enforced floor at 25% maximum efficiency loss).

760 *Data availability*

761 All simulation parameters, YAML configuration files, orchestration
762 scripts, and analysis code needed to reproduce every figure and table are
763 available from the corresponding author upon reasonable request and will
764 be deposited in a public repository with a DOI upon acceptance. The
765 PT-HOPS/SBD implementation is based on the open-source MesoHOPS
766 library (v1.7) [20].

767 *Code availability*

768 The digital-twin simulation framework (orchestrator, QAOA scheduler,
769 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond
770 pathogen sensor, and GQAS compliance checker) and all simulation pa-
771 rameters, including the 12-mode vibronic bath spectral density and the
772 8-site FMO electronic Hamiltonian specified in the machine-readable
773 `parameters.yaml` configuration file (v1.4), are available from the corre-
774 sponding author upon reasonable request and will be deposited in the same
775 public archive with a DOI upon acceptance.

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781 **CRedit authorship contribution statement**

782 **S.C. Tegua Kouam:** Conceptualization, Methodology, Software, For-
783 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
784 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
785 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
786 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
787 editing.

788 **Declaration of competing interest**

789 The authors declare that they have no known competing financial inter-
 790 ests or personal relationships that could have appeared to influence the work
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 800 was reviewed and verified by the authors, who take full responsibility for the
 801 content of the publication.

802 **Nomenclature**

	Symbol	Meaning
	Φ_{FT}	Forward (reaction-center) transfer yield
	τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coher- ence)
	λ_{res}	Residual reorganization energy of polaron-dressed states
	g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
	κ / Q	Plasmon linewidth / quality factor
803	$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
	Γ_{RC}	Reaction-center trapping rate
	$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
	$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shad- ing factor
	NEB	Net Ecological Benefit
	$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
	QBER	Quantum bit error rate (BB84 link)
	Δg	Groundwater gravity anomaly
	$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

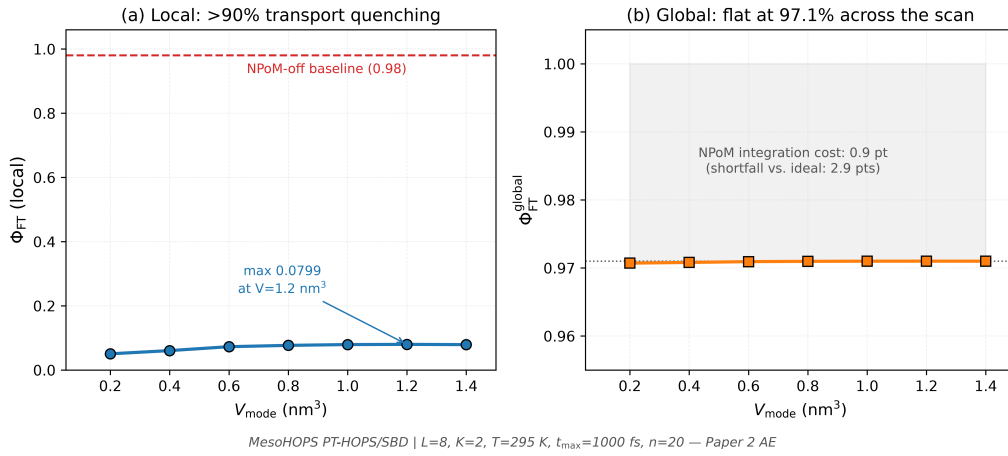


Figure 1: Local quenching, global stability: the central result in one picture. (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20$, $T = 295$ K): locally, transport is quenched by more than 90 % relative to the pristine FMO baseline (0.98), with a factor-1.6 variation across the accessible range and a maximum of 0.0799 at $V_{mode} = 1.2$ nm³ (Table 2). (b) Area-weighted global canopy trapping yield for the 1 % sentinel fraction (Equation (5)): flat at 0.971 across the entire scan — the deep local suppression is absorbed at a constant 0.9-point canopy-scale cost (0.980 to 0.971), leaving the canopy 2.9 points below an ideal yield of 1.0. The flat curve in (b), not the quenching in (a), is the design-relevant outcome: diagnostics integrate into the stack without compromising productivity.

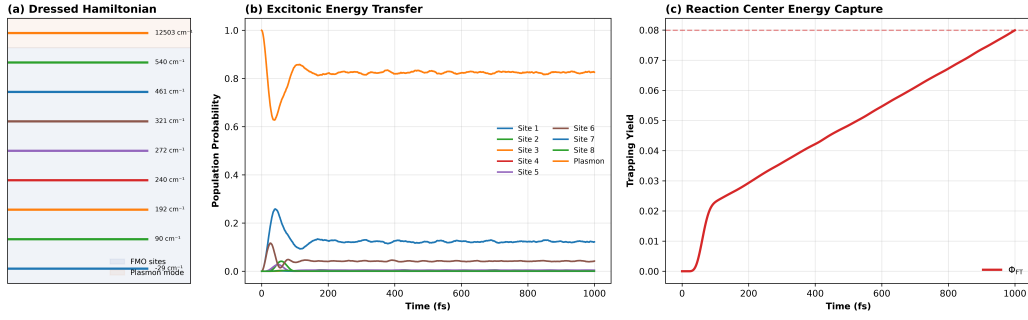


Figure 2: Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin. (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Population dynamics of the $n = 20$ production run at $V_{\text{mode}} = 1.2 \text{ nm}^3$ and 295 K (the scan optimum of Table 2): the plasmon mode (Site 9) loads within the first 10 fs and captures 83 % of the excitation, starving the trapping sites — the microscopic origin of the 91.8% local penalty. (c) Cumulative reaction-center trapping yield $\Phi_{\text{FT}}(t)$ for the same run, saturating at 0.0799. All data: $L = 8$, $K = 2$, $n = 20$ disorder realizations (Section S11). The passive-fraction dynamics underlying the coherence and yield gains of Figure 2c are reported in [25]. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield.

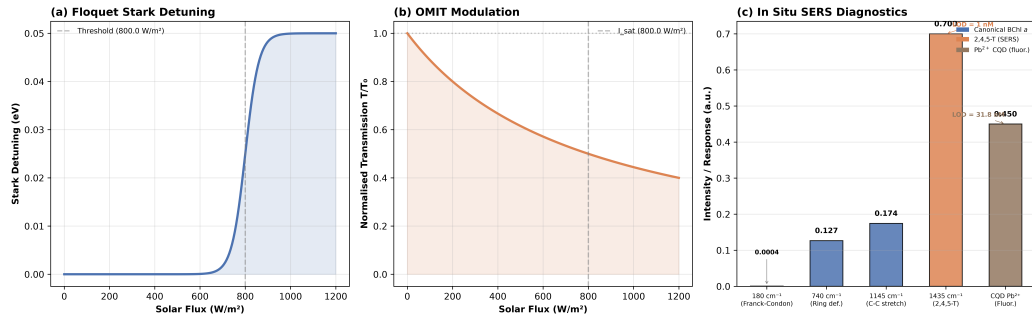


Figure 3: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux: the sigmoidal switch detunes the excitonic levels above 800 W m^{-2} , cutting energy transfer to FMO by up to 40 % — the solid-state analogue of non-photochemical quenching. (b) OMIT transmission modulation at 750 nm and 820 nm: at 1000 W m^{-2} the two filter pass-bands themselves close to 44 % of baseline, protecting the exciton budget precisely where the co-designed filter concentrates it. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

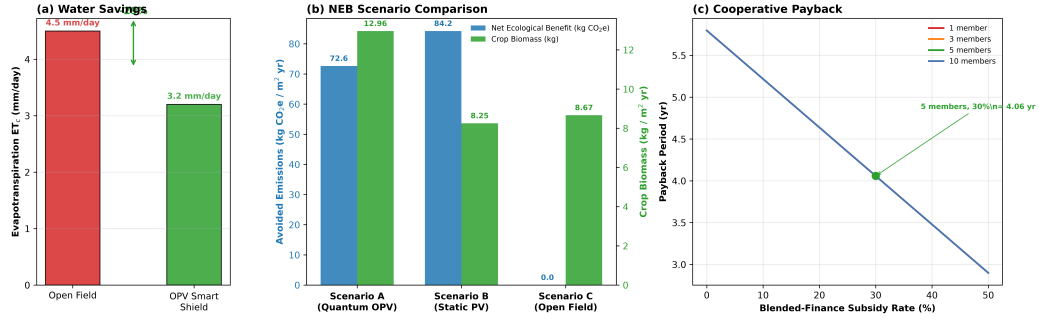


Figure 4: **Water-Energy-Food network assessment.** (a) FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. (b) Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $72.6 \text{ kg}_{CO_{2e}} \text{ m}^{-2} \text{ yr}^{-1}$ with an effective crop biomass of $13.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $84.2 \text{ kg}_{CO_{2e}} \text{ m}^{-2} \text{ yr}^{-1}$ but only $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The comparison is deliberately uncomfortable: an opaque static module still buys more carbon than the quantum shield, but it does so by suppressing the canopy to 84% of reference biomass, whereas the quantum shield preserves near-baseline productivity (0.971 global trapping yield, 13.0 including biostimulation) — exactly the food-energy trade-off that motivates spectral co-design. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. (c) Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.32 yr (net annual cash-flow of $26\,090 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modelled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

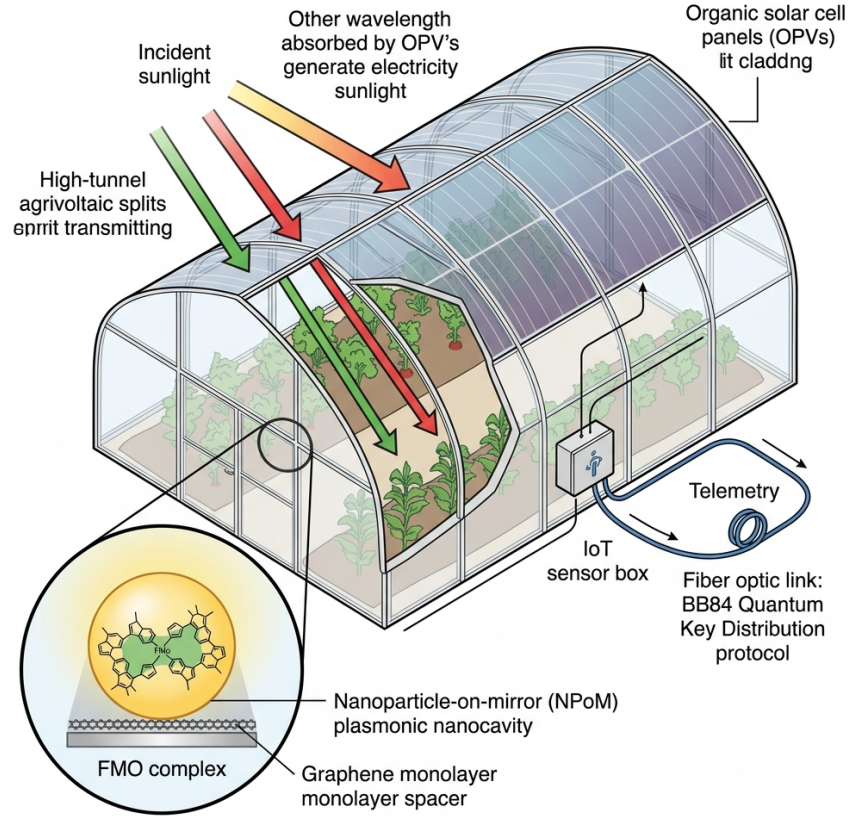


Figure 5: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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